Pascal's Triangle (1641)

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

- **Corollary**

Pascal's identity, together with the initial conditions $\binom{n}{0} = \binom{n}{n} = 1$ for all integers *n*, can be used to give a geometric arrangement of the binomial coefficients in a triangle, called Pascal's triangle, as follows:

$\binom{0}{0}$

$\binom{1}{0}\ \binom{1}{1}$

$\binom{2}{0}\ \binom{2}{1}\ \binom{2}{2}$

$\binom{3}{0}\ \binom{3}{1}\ \binom{3}{2}\ \binom{3}{3}$

$\binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4}$

$\binom{5}{0}\ \binom{5}{1}\ \binom{5}{2}\ \binom{5}{3}\ \binom{5}{4}\ \binom{5}{5}$

$\binom{6}{0}\ \binom{6}{1}\ \binom{6}{2}\ \binom{6}{3}\ \binom{6}{4}\ \binom{6}{5}\ \binom{6}{6}$

$\binom{7}{0}\ \binom{7}{1}\ \binom{7}{2}\ \binom{7}{3}\ \binom{7}{4}\ \binom{7}{5}\ \binom{7}{6}\ \binom{7}{7}$

$\binom{8}{0}\ \binom{8}{1}\ \binom{8}{2}\ \binom{8}{3}\ \binom{8}{4}\ \binom{8}{5}\ \binom{8}{6}\ \binom{8}{7}\ \binom{8}{8}$

...

(a)

By Pascal's identity:

$\binom{6}{4} + \binom{6}{5} = \binom{7}{5}$

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

1 7 21 35 35 21 7 1

1 8 28 56 70 56 28 8 1

...

(b)

